Local Search Part 2

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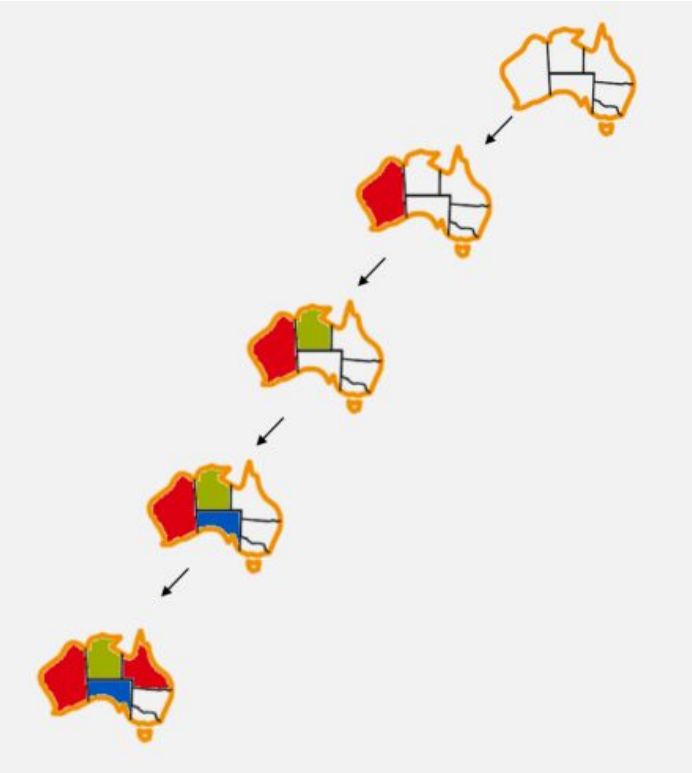
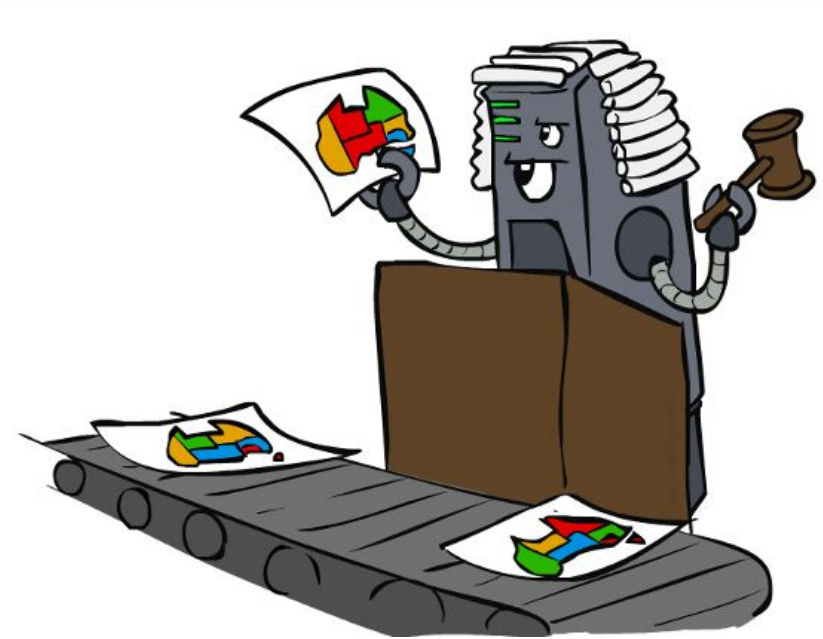
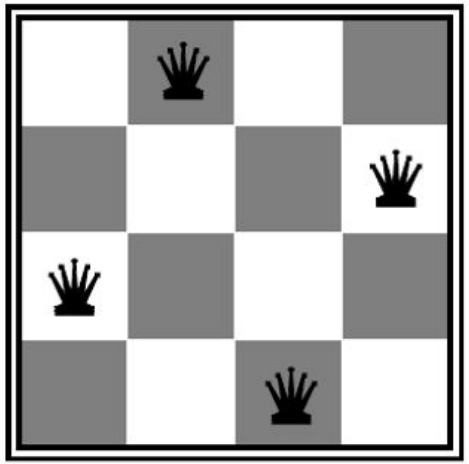
Some of Slides have been adopted from Berkeley CS188

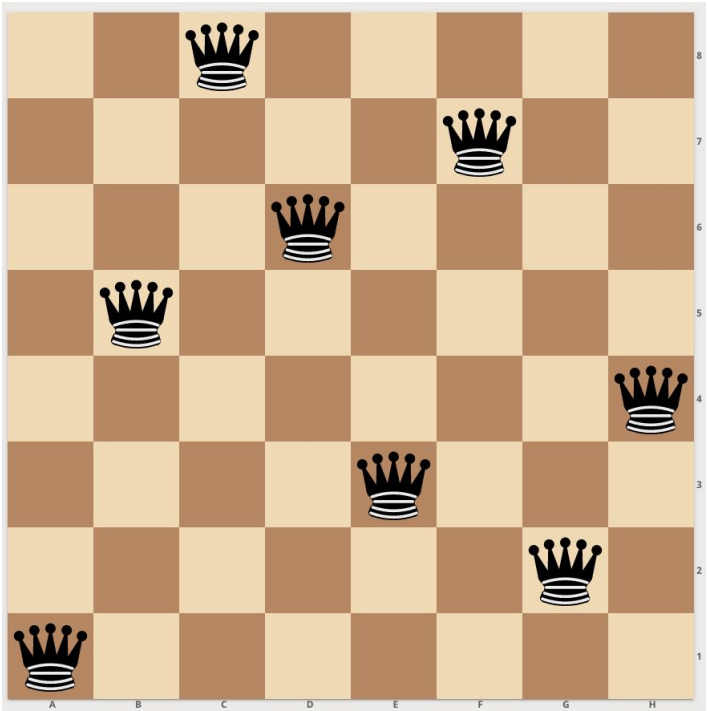
Partial State VS Complete State





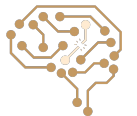
Example: N-Queens Problem





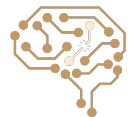
The Twist

- Goal states are complete states.
- All complete states exist at the lowest level of the search tree.
- Every path to a goal state is considered optimal (path optimality is irrelevant).
- Previous concepts (A*, heuristics, etc.) no longer apply.
- Question: Why not just search exclusively among complete states?
- Result: Tree searching is no longer necessary.



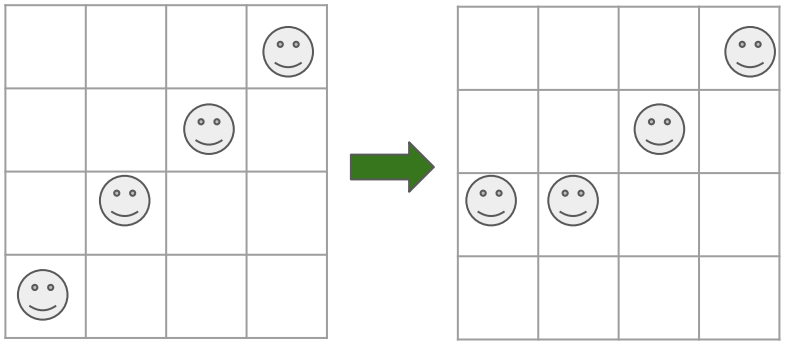
Local Search

- State Space: All complete configurations (complete states)
- State State: A random complete state
- The local search algorithm is just an iterative improvement: Keep a single "current" state, try to improve it by going to a neigbor state
- Local search improves a single option at each time (no frontier)
- Memory-less: only remember current state



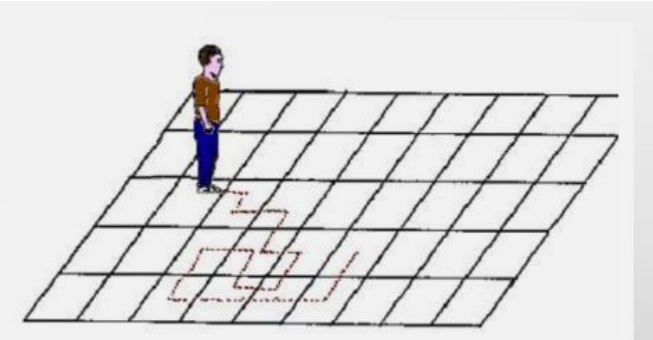
Neighboring in Local Search

- Neighbors of a state: States that are different in only one variable



- Question: Which neighbor must be choose?

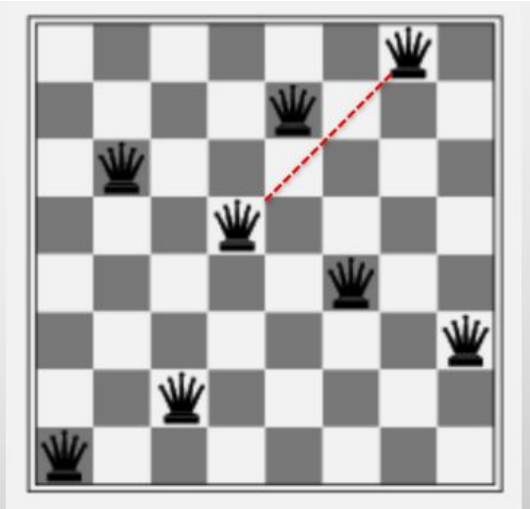
Random Walk



- Randomly pick a neighbor of the current state
- Random Walk algorithm is asymptotically complete (as the number of iterations increases, the probability that the algorithm has found the goal state approaches 1)

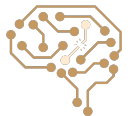
Fitness Function

- How good is each state?
- For N-Queens: # pairs of incompatible queens
- What about map coloring problem?

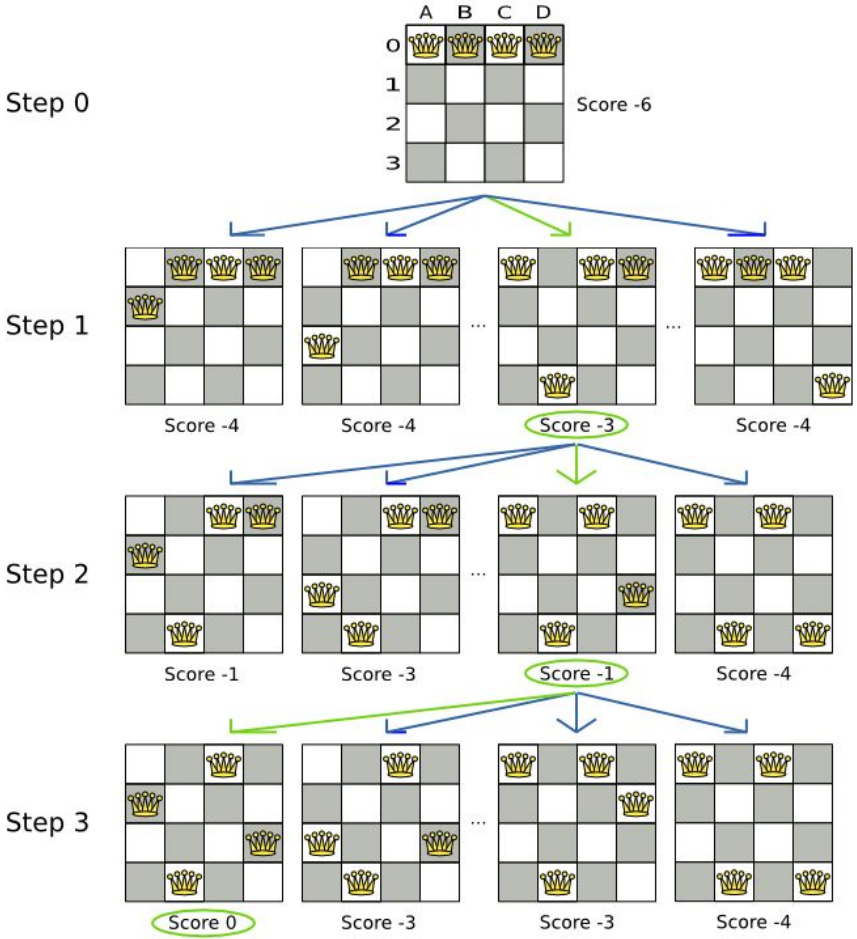


Hill-climbing

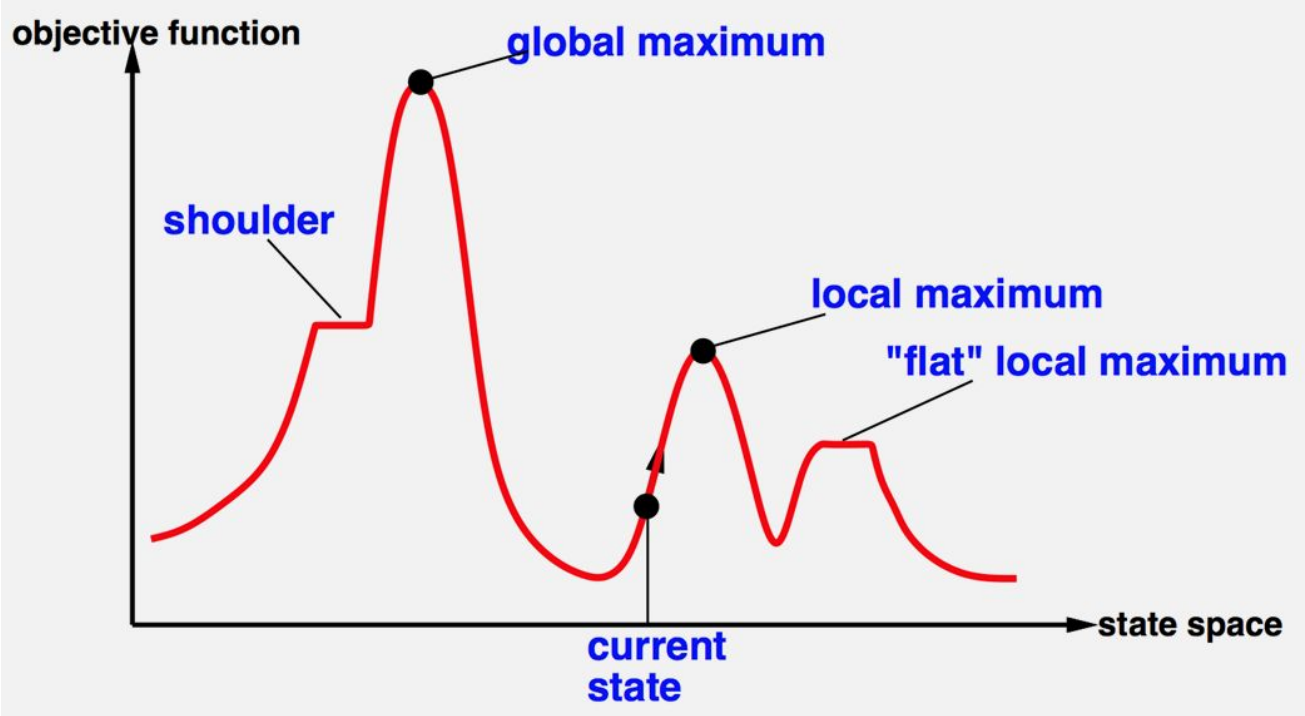
- Start from a random complete state
- Repeat: move to the best neighbor that is better than current state
- If no neighbors better than current state, quit
- Also called steepest-ascent hill climbing
- Complete?



Hill-climbing on N-Queens

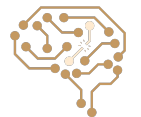


Hill Climbing Hells



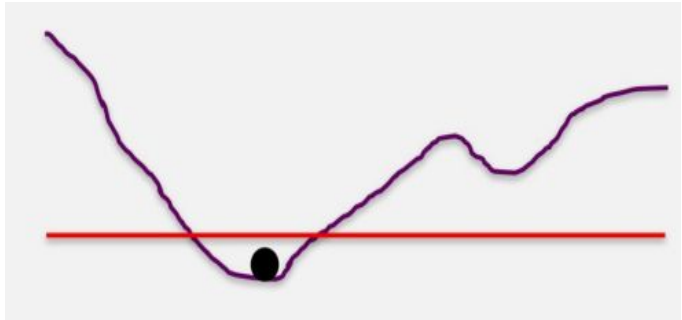
Hill-climbing Variations

- + Limited Number of Consecutive Sideway Moves
- + Tabu List (a fixed length queue)
- + Random Walk
- + Random Restarts



Simulated Annealing (SA)

- Idea: escape local maxima by allowing some “bad” moves
- but gradually decrease their size and frequency
- Imagine letting a ball roll downhill on the function surface
- Now shake the surface, while the ball rolls
- Gradually reducing the amount of shaking



Simulated Annealing (SA) Algorithm

```
function Simulated-Annealing(problem, schedule) returns solution state
current ← Make-Node(Initial-State[problem])
for t ← 1 to infinity
  T ← schedule[t]
  if T = 0 then return current
  next ← Random-Successor(current)
  ΔE ← f-Value[next] - f-Value[current]
  if ΔE > 0 then current ← next
  else current ←next with probability e^(ΔE/T)
end
```

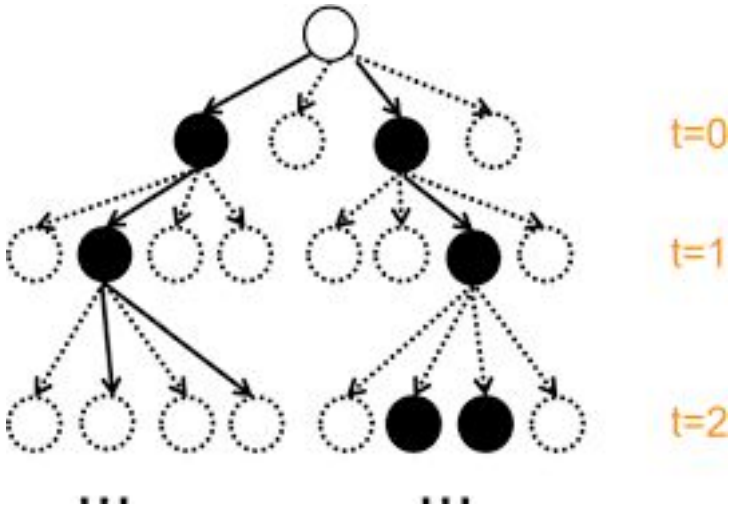
- Pick a random successor of the current state
- If it is better than the current state go to it
- Otherwise, accept the transition with a probability

$T(t) = schedule[t]$ is a decreasing series

Local Beam Search

- Instead of just one in hill-climbing and simulated annealing
- Keep track of k states simultaneously
  - Initially: k randomly selected states
  - Next: determine all neighbors of all k states
  - If any of successors is goal: finish
  - Else select k best from the all neighbors and repeat
- Not same as k independent random-start searches run in parallel!

Local Beam Search Example

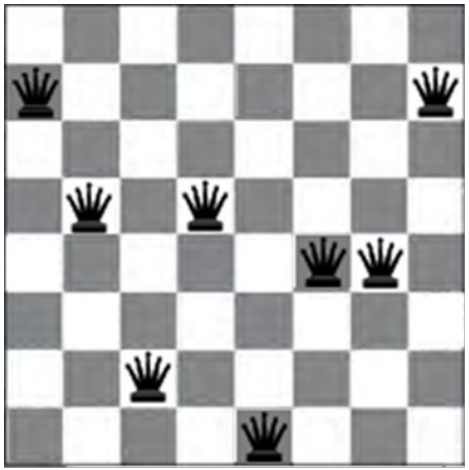


Genetic algorithms

- No neighboring concept
- Start with k randomly generated states (initial population)
- A state is represented as a string over a finite alphabet (Chromosome), often a string of 0s and 1s
- Evaluation function (fitness function)
- Produce the next generation of states by **selection**, **crossover**, and **mutation**

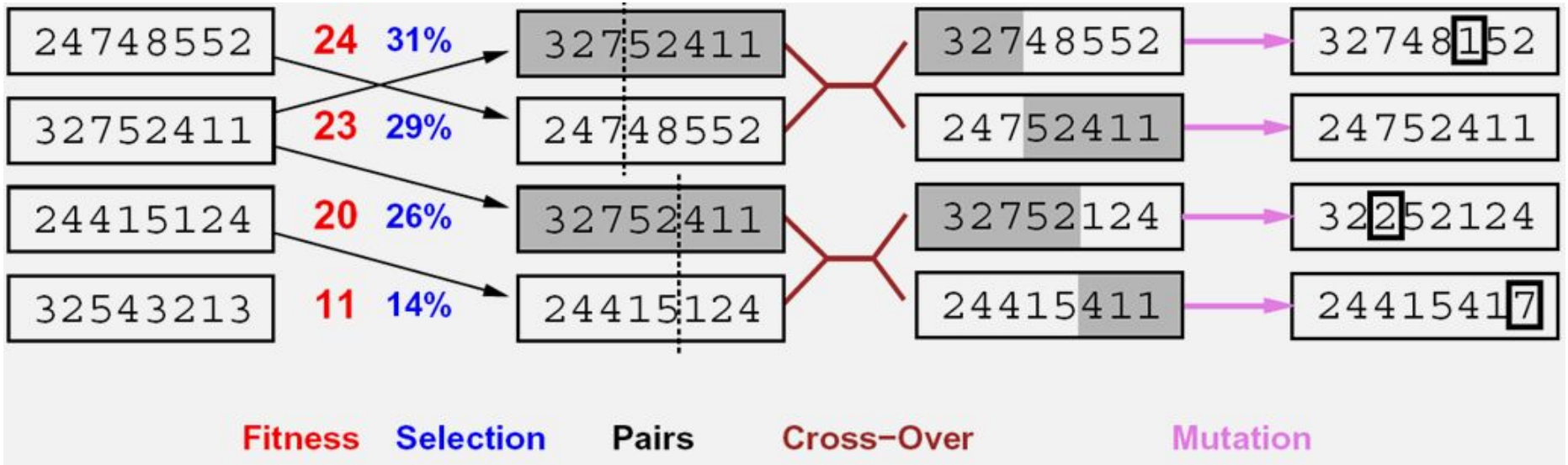
Chromosome & Fitness: 8-queens Problem

- Describe the individual (or state) as a string

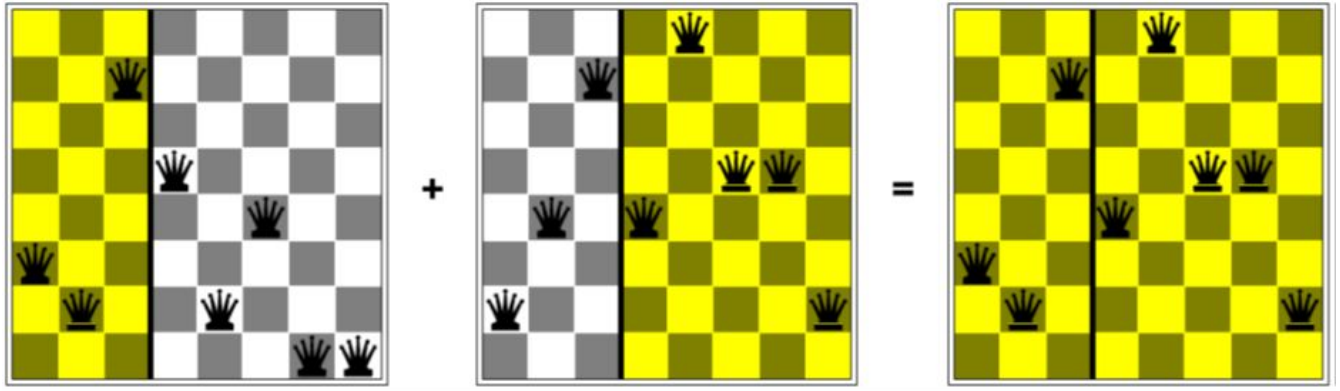


- Fitness function: number of non-attacking pairs of queens (24 for above figure)

Genetic algorithms

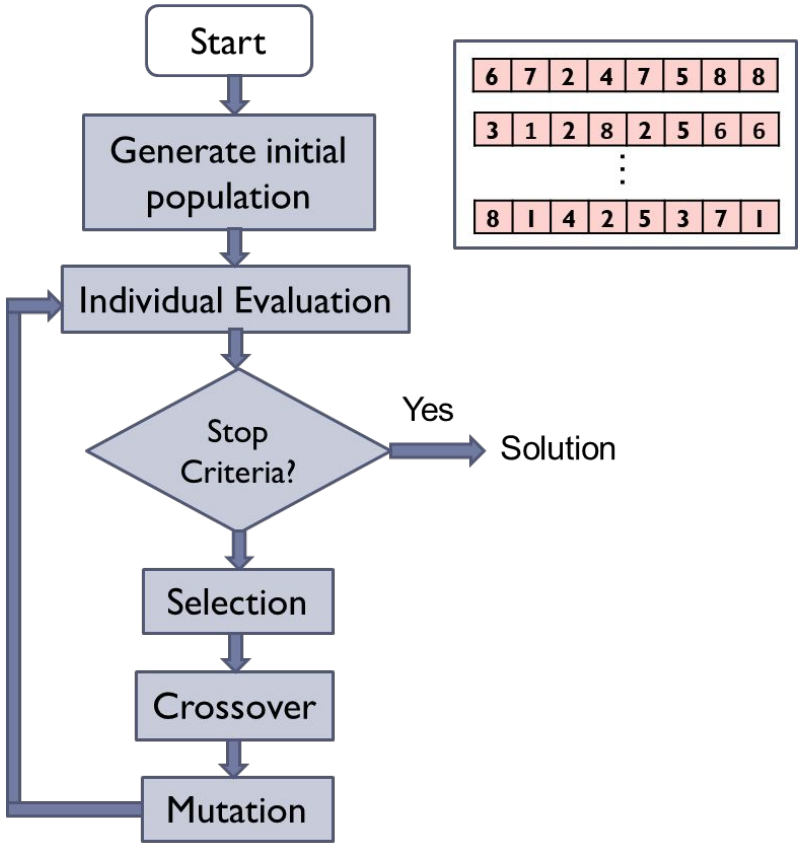


Example: N-Queens

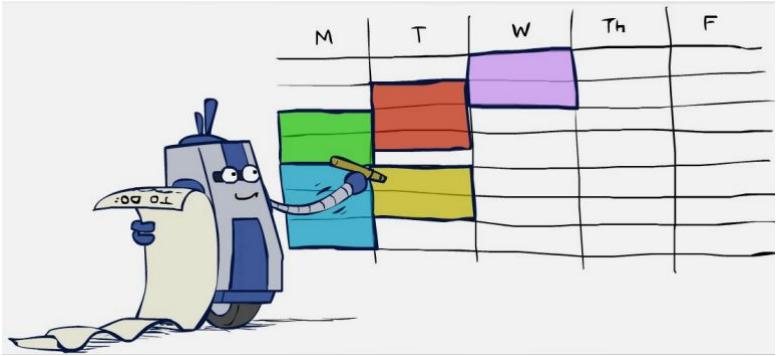


- Does crossover make sense here?
- What would mutation be?
- What would a good fitness function be?

Genetic Algorithm Diagram



Next Session: Constraint Satisfaction Problem (CSP)

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